Shortlex Enumerator and Recursive Languages

Problem

Recall the shortlex enumerator from class. Prove that the language recognized by any shortlex enumerator is recursive.

Solution

A shortlex order lists strings first by length and then lexicographically among strings of the same length. For example: ε, a, b, aa, ab, ba, bb, ... (depending on the alphabet). A shortlex enumerator for a language L outputs the strings of L in this order.

Suppose a Turing machine E enumerates a language L in shortlex order. Because the enumeration is strictly increasing in this order, once a string greater than w appears, the string w can never appear later.

To decide whether a string w belongs to L, we simulate the enumerator E:

1. Run the enumerator step by step. 2. Whenever the machine outputs a string u: - If u = w, accept. - If u is greater than w in shortlex order, reject. 3. Otherwise continue.

If $w \in L$, the enumerator eventually prints w and the algorithm accepts. If $w \notin L$, the enumerator eventually outputs a string larger than w in shortlex order, after which w cannot appear, so the algorithm rejects.

This procedure halts on every input and correctly decides membership in L. Therefore the language recognized by any shortlex enumerator is decidable (recursive).